

E9 205 Machine Learning for Signal Processing

**ML, MAP, MMSE and Gaussian
Modeling**

04-09-2017

Recap ...

- ❖ Decision Theory
 - ❖ Inference problem
 - ❖ Finding the joint density $p(\mathbf{x}, \mathbf{t})$
 - ❖ Decision problem
 - ❖ Using the inference to make the classification or regression decision

Decision Problem - Classification

- ❖ Minimizing the mis-classification error
- ❖ Decision based on maximum posteriors

$$\operatorname{argmax}_j p(C_j|\mathbf{x})$$

- ❖ Loss matrix
 - ❖ Minimizing the expected loss

$$\operatorname{argmax}_j \sum_k L_{k,j} p(C_k|\mathbf{x})$$

Approaches for Inference and Decision

I. Finding the joint density from the data.

$$p(C_k|\mathbf{x}) \propto p(\mathbf{x}|C_k)p(C_k)$$

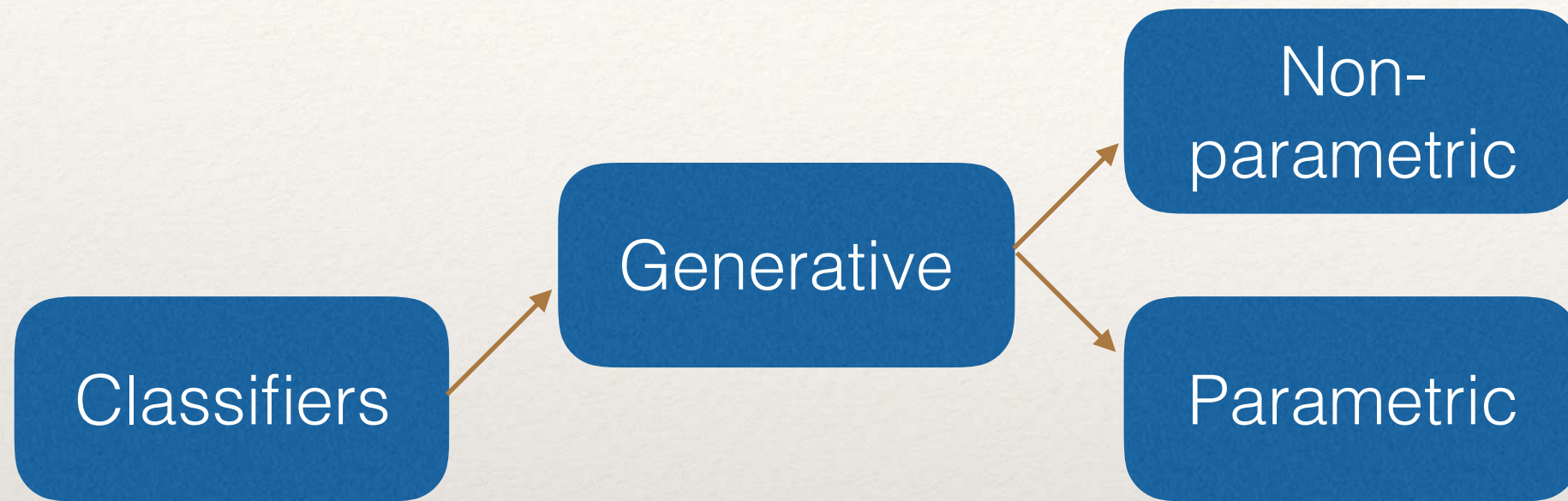
II. Finding the posteriors directly.

III. Using discriminant functions for classification.

Decision Rule for Regression

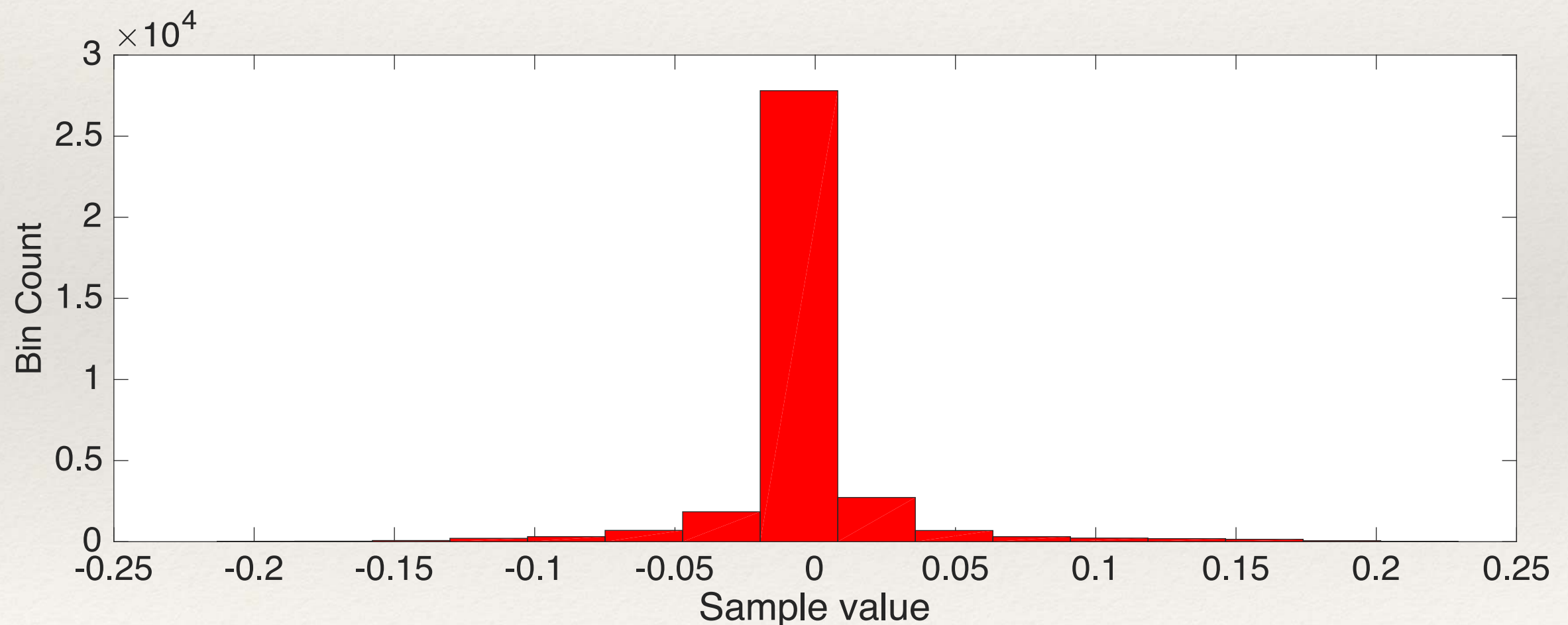
- ❖ Minimum mean square error loss
- ❖ Solution is conditional expectation.

Generative Modeling



Non-parametric Modeling

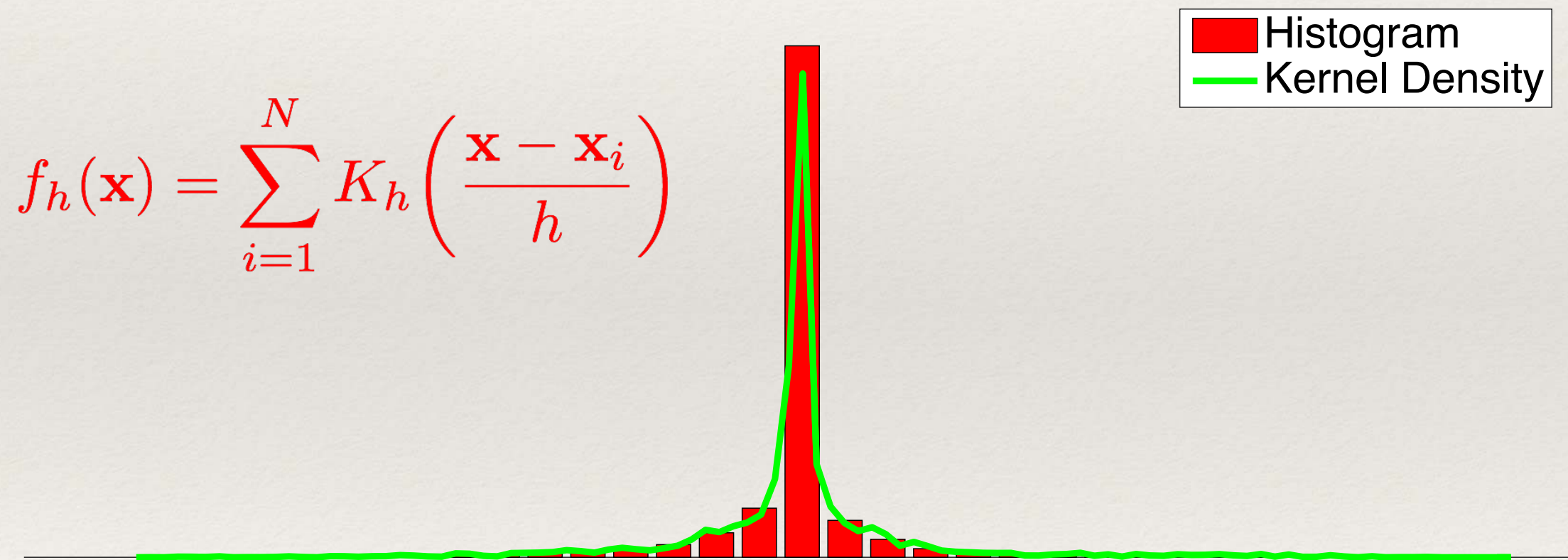
- **Non-parametric** models do not specify an apriori set of parameters to model the distribution. Example - Histogram



The density is not smooth and has block like shape.

Non-parametric Modeling

- **Non-parametric** models do not specify an apriori set of parameters to model the distribution.
 - Example - Kernel Density Estimators



Kernel is a smooth function which obeys certain properties

Non-parametric Modeling

- Non-parametric methods are dependent on number of data points
 - Estimation is difficult for large datasets.
- Likelihood computation and model comparisons are hard.
- Limited use in classifiers

Parametric Models

- ❖ Collection of probability distributions which are described by a finite dimensional parameter set

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_K) \quad P = \{P_{\boldsymbol{\theta}}\}$$

- Examples -

- Poisson Distribution

$$p_{\lambda}(j) = \frac{\lambda^j}{j!} e^{-\lambda}$$

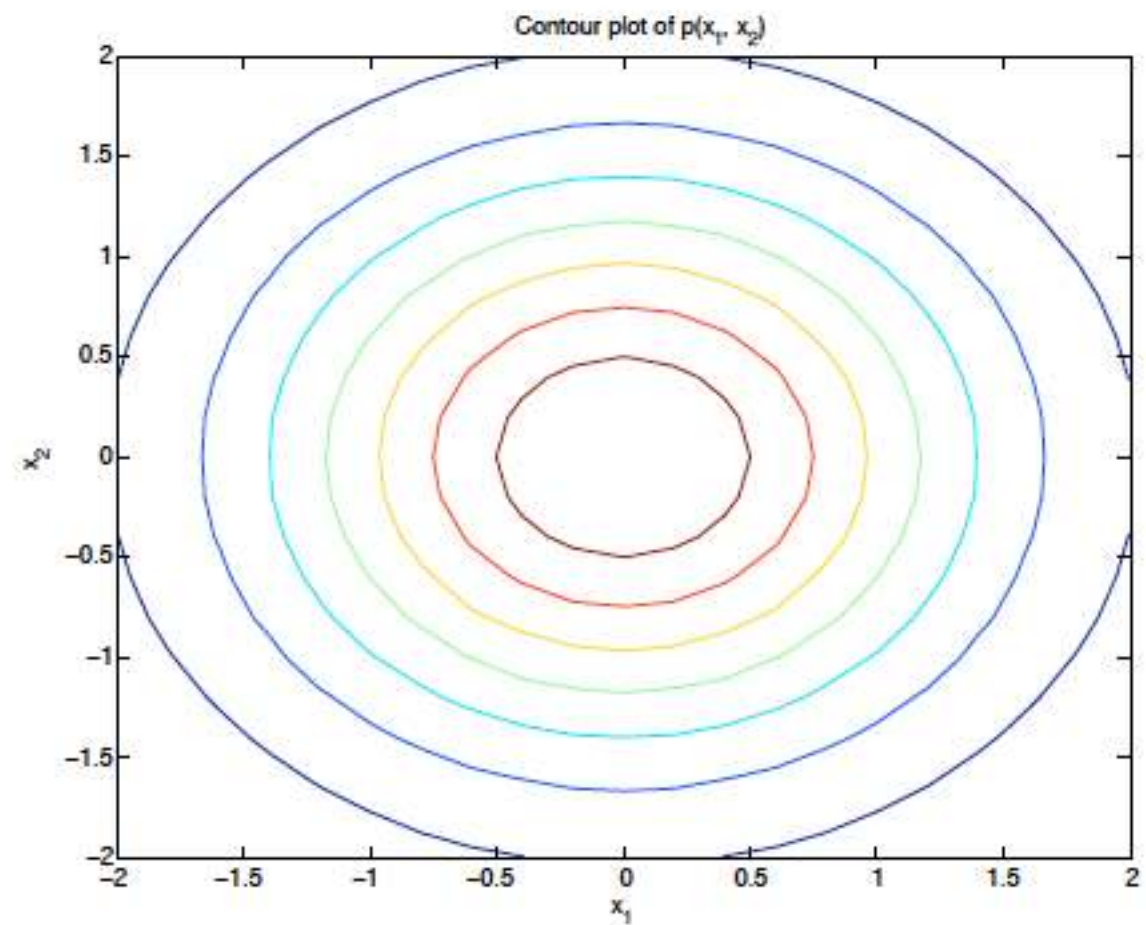
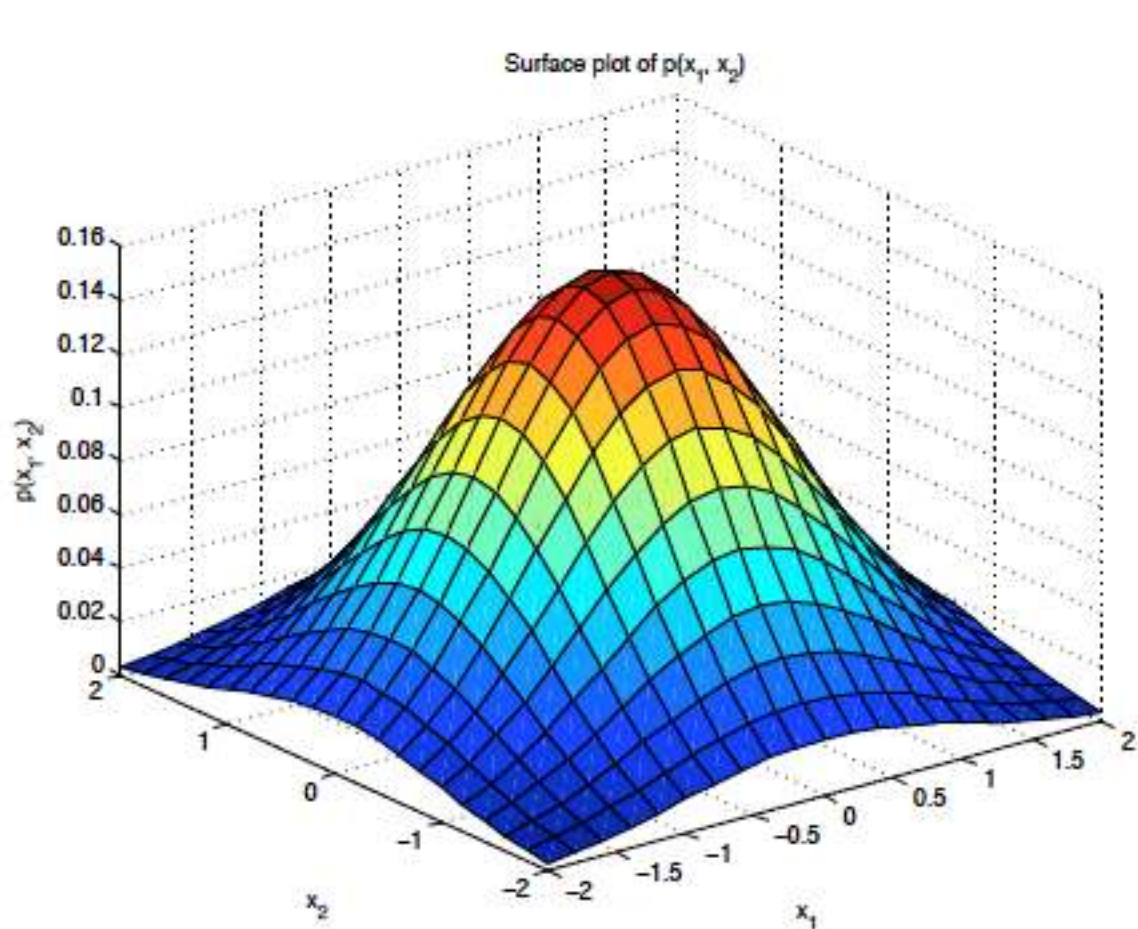
- Bernoulli Distribution

$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{i=1}^D \mu_i^{x_i} (1 - \mu_i)^{1-x_i}$$

- Gaussian Distribution

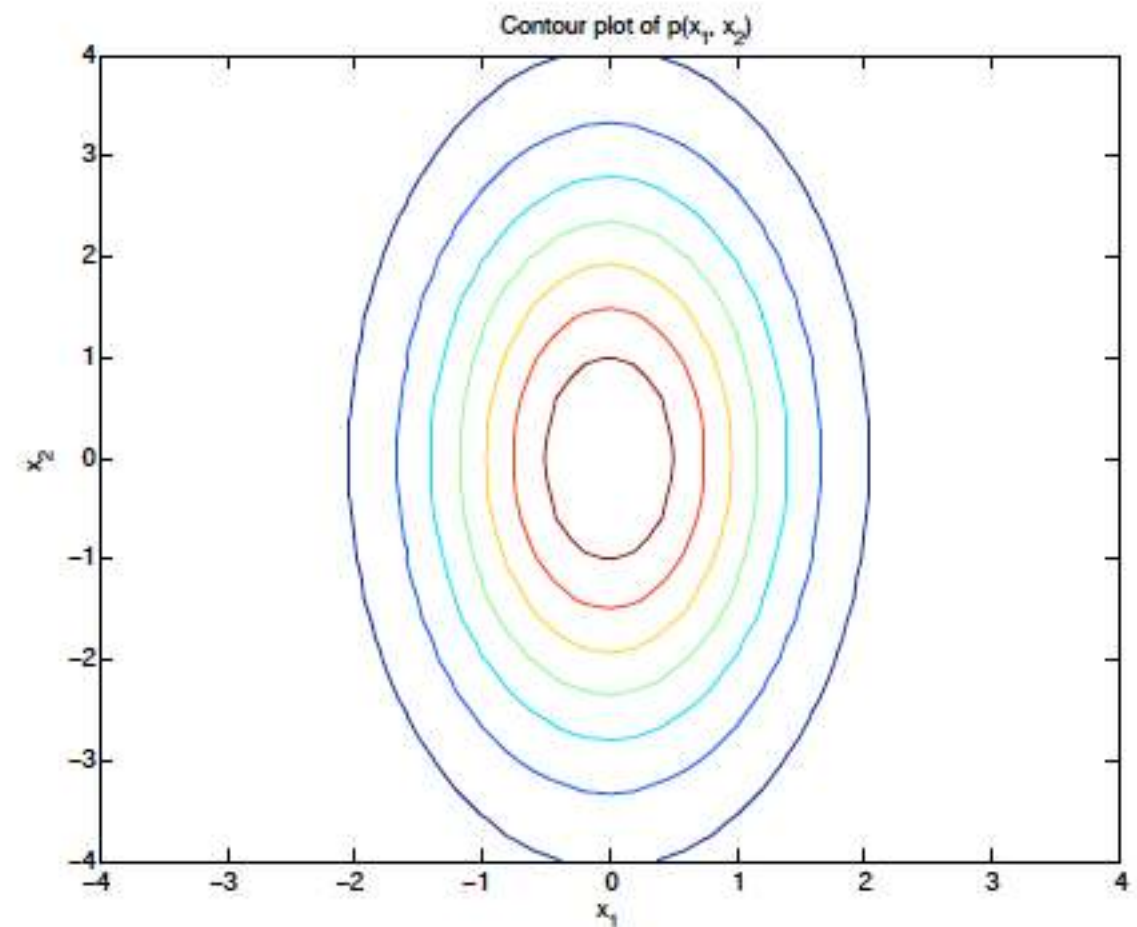
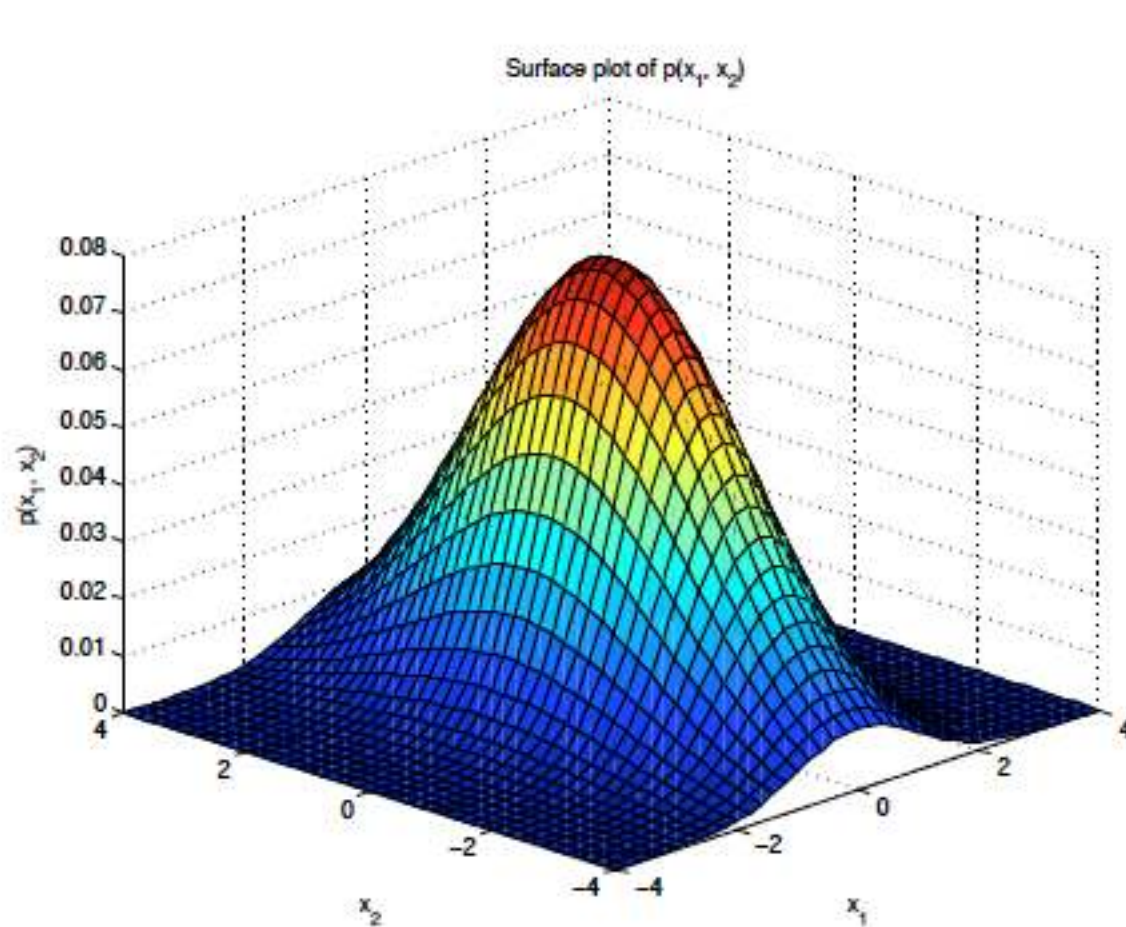
$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp\left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}$$

Gaussian Distribution



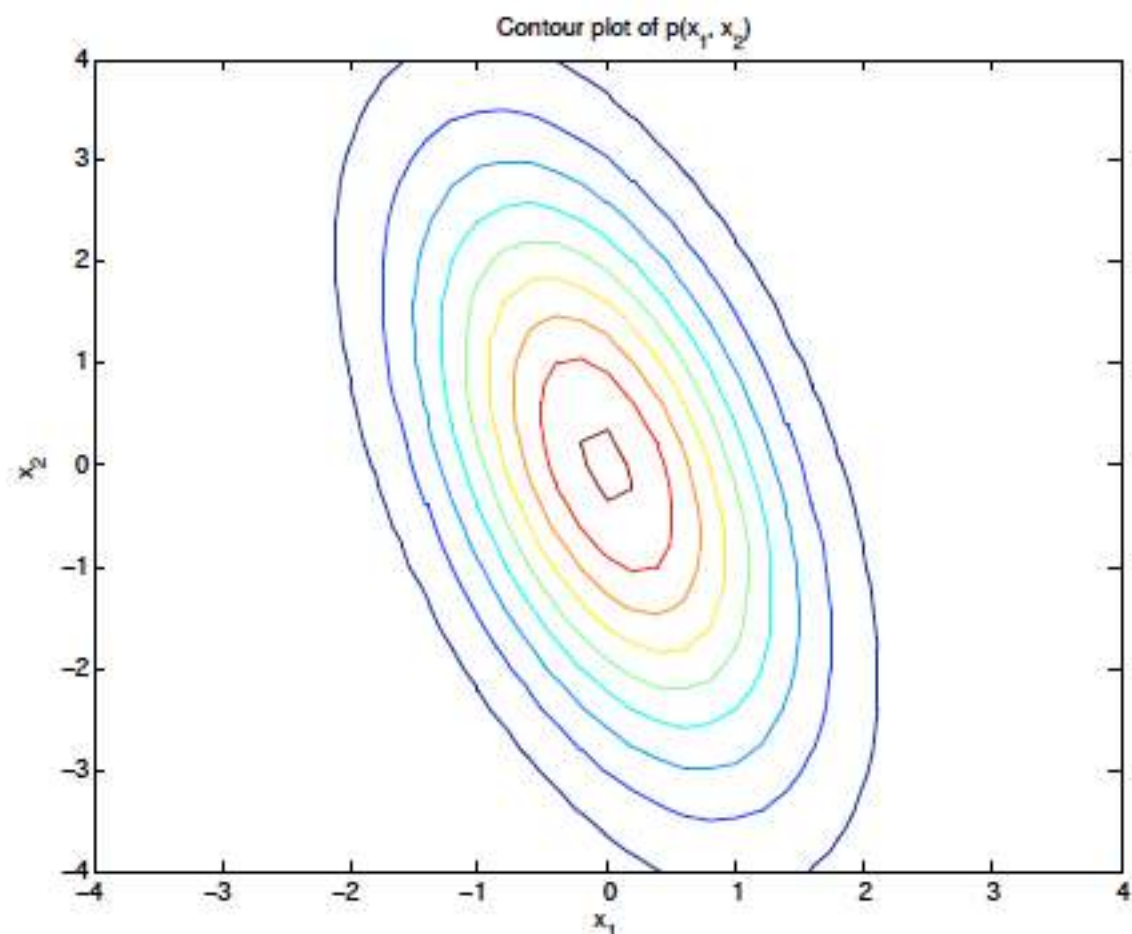
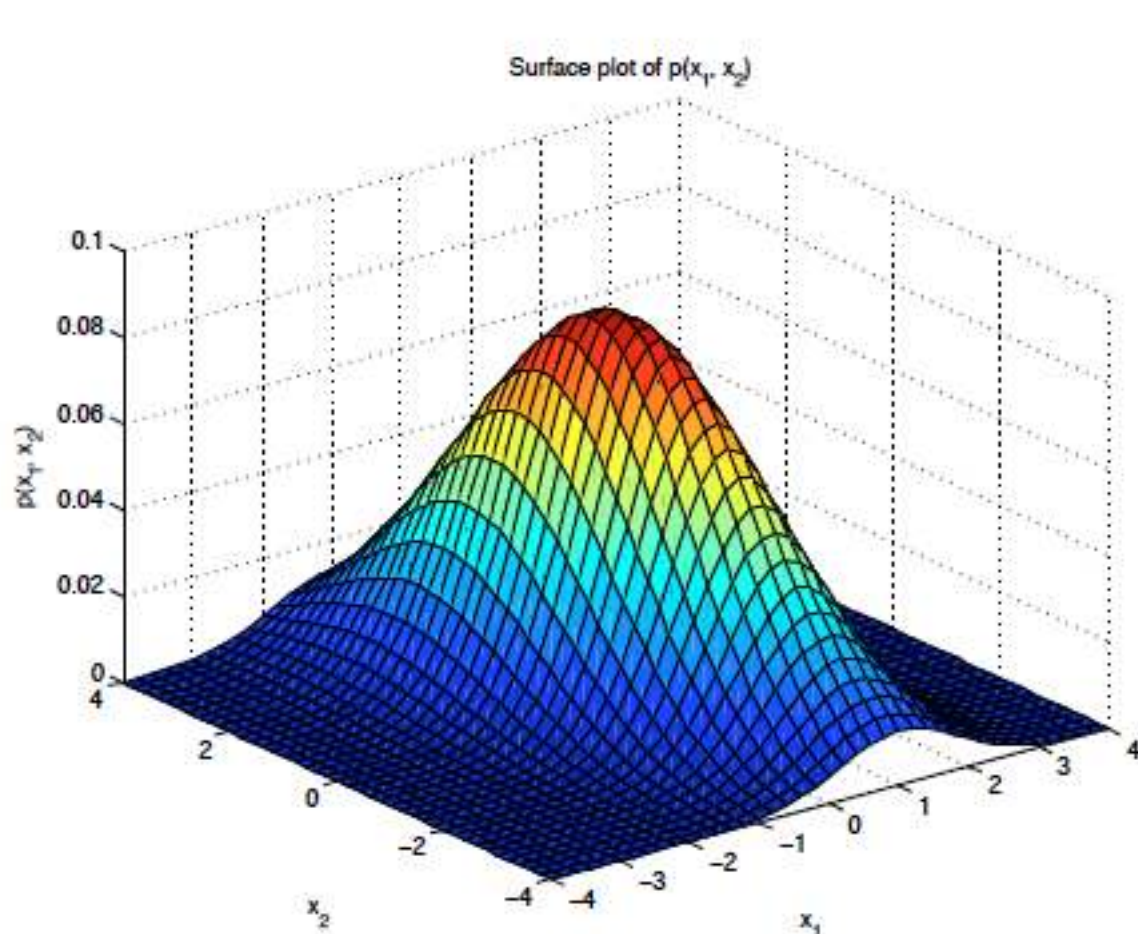
Points of equal probability lie on on contour
Diagonal Gaussian with Identical Variance

Gaussian Distribution



Diagonal Gaussian with different variance

Gaussian Distribution



Full covariance Gaussian distribution